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From Concept to Reality: A Systematic Review of Urban Digital Twin and Their Applications in Smart Cities

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Abstract. Metropolises comprise six primary systems that serve distinct domains: governance, environment, mobility, economy, living, and society. These systems work together to create a macrosystem that constitutes a city. In contemporary urban planning, Smart Cities (SCs) have emerged as a means of effectively managing and integrating critical infrastructure, optimizing resource allocation, coordinating maintenance activities, and addressing complex urban challenges while prioritizing security. Over the past few years, cutting-edge technologies like Wireless Sensor Networks, Building Information Modeling, Internet of Things (IoT), Virtual Reality, Big Data Analytic, Artificial Intelligence (AI), Cloud Computing, Blockchain, and Cyber-Physical Systems have paved the way for intelligent and groundbreaking solutions in a variety of industries. Built on the knowledge gained from developing SCs and implementing Digital Twins (DTs) in other sectors, the Urban Digital Twin (UDT) has rapidly expanded in just a few years. UDT technology utilizes real-time data, simulation, machine learning, and reasoning to provide valuable assistance in decision-making processes. However, more than informatization and digitalization practices are required to embrace UDT fully. There has yet to be a comprehensive analysis of how to apply these features of DTs to manage SCs better. This review paper analyzed 150 papers that focused on DT-supported SCs. We aim to identify the drivers, challenges, and future directions for applying UDT-supported technologies. We systematically investigated the characteristics and mechanisms of DTs, categorized different technologies based on their application stages, and identified the key challenges, benefits, and open issues. The findings inform stakeholders in urban management to establish a foundation for future design, development, and widespread adoption of UDT.

Keywords: Digital Twins · Smart Cities · Urban Management · Internet of Things (IoT) · Decision-making

1 Introduction

The 21st century has experienced clashes between extensive and rapid urbanization along with delayed urban construction, resulting in significant infrastructure and capacity challenges for major urban centres and smaller communities. Urban planners and scholars

have sought innovative approaches to improve urban governance in complicated urban systems and address existing social problems in city management. Governments and enterprises increasingly use the concept of Smart Cities (SCs) to enhance urban environments by implementing digitization technology. An Urban Digital Twin (UDT) is a technical approach that leverages cutting-edge technologies, including the Internet of Things (IoT), big data management, cloud computing, and virtual reality, to create an instantaneous, real-time digital replicate of a city [1, 2]. A digital twin (DT) is a virtual representation of a real-world object or system that collects data from many sources, such as IoT sensors, building information modelling (BIM) models, geographic position data, and other data sources. It uses this data to emulate the behaviour and characteristics of the physical asset or system by creating a digital counterpart. This system can simulate, predict, explore, and visualize data via Machine Learning (ML) and AI technologies. It offers in-depth insights into the system's functioning [3], which has significant consequences for city planning, management, and decision-making [4, 5].

Although the implementation of UDT is expected, several obstacles must be overcome before it becomes a reality. These obstacles include technical, ethical, and practical challenges. One major challenge is the collection, storage, and processing of a vast amount of diverse data necessary for developing comprehensive decision trees encompassing all aspects of a city [3, 6]. Additionally, integrating data privacy and security into the platform presents substantial challenges that need to be addressed for the broad acceptance and successful implementation of DTs [2, 7]. Hence, there is a pressing need for a holistic investigation and examination of DT systems across different domains of SCs. This study thoroughly examines the fundamental structure and diverse uses of UDT. The article comprehensively analyzes the relevant literature from the past 20 years. In the beginning, the article provides clear definitions of the concepts of DT and SC. It also compares and summarizes the system development of UDT. Furthermore, it classifies and summarizes the technical applications of DT across the entire data management process. Additionally, this article discusses the benefits, challenges, and open issues associated with existing DTs. Finally, it offers insights into the future of DT research and the upcoming direction of DT research.

2 Conceptualizing Digital Twin and Smart Cities

2.1 The Evolution Trend of Digital Twin System Development

The term “digital twin” is widely credited to Dr. Michael Grieves, who first coined the phrase in his 2002 presentation at the University of Michigan [8]. Grieves developed the concept as part of product lifecycle management, describing the DT as a mirror or shadow of the physical object created for simulation and analysis. NASA was one of the first organizations to apply DT technology by creating a ground-based duplicate of spacecraft for simulation and study purposes [9, 10]. This enabled predictive maintenance and failure prediction, improving spaceship safety and dependability. Following aerospace, the manufacturing industry embraced DTs for product design and process optimization with the emergence and development of “ubiquitous computing” and the implementation of IoT-related technologies. “Ubiquitous Computing”, proposed by Mark Weiser in 1988, provides a rich, responsive, and intelligent framework upon which DTs can be

built and operated, greatly enhancing their capabilities and applications across various domains [11]. IoT provides the continuous data stream and connectivity that enable DTs to offer real-time, accurate, and predictive insights for improved monitoring, control, and optimization of physical systems across various industries [12]. According to Jones and his colleagues, the predominant application of DTs is currently observed in the manufacturing sector [13]. Siemens, the leading industrial manufacturing company in Europe and a trailblazer in digital technologies, has created digital models for its gas turbine and compressor business. These models, known as Agent-Based Turbine Operations and Maintenance (ATOM), are a significant development. ATOM can effectively showcase maintenance operations and logistical procedures by processing multiple real-time data and simulating various scenarios. Visualizing the outcome might guide the organization in making more dependable investment choices [14]. General Electric Company developed a DT specifically for the GE90 engine to predict the deterioration of aircraft engines. This DT focuses on the engine's fan blades and is a virtual replica of the long-haul Boeing 777 aircraft. By accurately timing maintenance activities, the DT helps prevent issues such as fan blade stripping caused by adverse environmental conditions [15]. As DT was listed as one of Gartner's top 10 technology trends in 2017 [16], large software and industrial firms have incorporated DT in their offerings. The forthcoming trajectory of DT development is anticipated to increasingly converge towards creating adaptive autonomous twins. Without human intervention, these twins can autonomously adapt their activities based on changing environmental circumstances and data inputs [17]. Figure 1 illustrates the 50-year development trend of DT from the 1970s to the present.

Originally, DT applications focused on individual elements and broadened to include complete systems. DTs have become a crucial component of Industry 4.0, which signifies the fourth wave of industrial transformation. They facilitate the integration and interoperability of diverse systems, providing real-time visibility and predictive capabilities. As a result, DTs enable more streamlined and adaptable processes [18]. Tao and his colleagues comprehensively examined the utilization of DTs in product lifecycle management. They highlighted the shift from static 3D models to dynamic systems that integrate real-time data feeds, predictive analytics, and decision-making support [19]. The transition above has profoundly affected multiple aspects of product creation, encompassing design, production, and post-service stages. Incorporating DTs into the Industrial Internet of Things (IIoT) and big data analysis has significantly improved the capacity for immediate monitoring and anticipatory maintenance [20].

The Virtual Singapore project might be regarded as the initial implementation of DTs in the SC domain. This project aimed to provide a dynamic 3D model of the city and a collaborative data platform to facilitate urban planning, research, and public services [21, 22]. With the rise of IoT, big data, and AI, UDTs have emerged as a novel urban planning and management approach, allowing advanced analytics, modelling, and optimization through virtual representations [23], especially for complex urban environments [24].

2.2 Architecture of UDT and Supported Applications

Several architectural concepts have been proposed for UDTs. In Grieves' publication [25], a white paper on DT, it is stated that the fundamental DT model comprises three

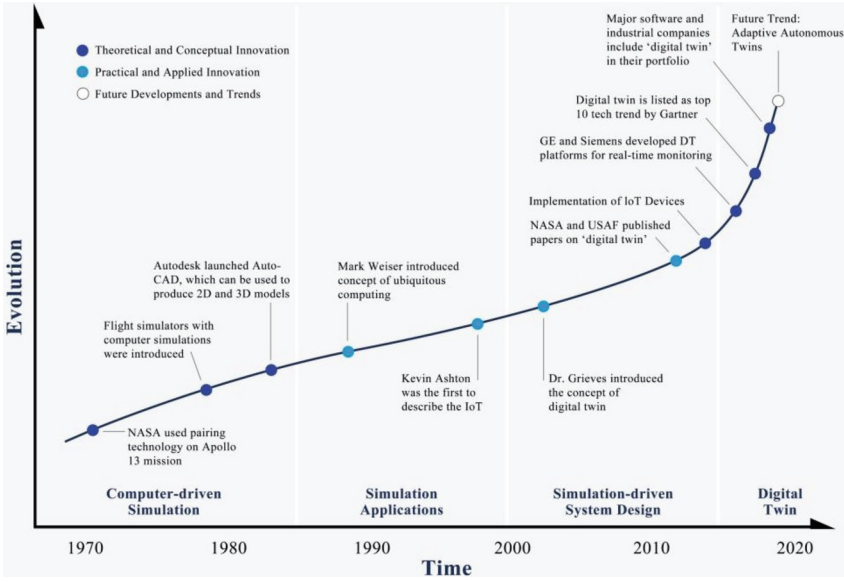


Fig. 1. The developing trend of the digital twin system.

primary components: (a) tangible items in the physical realm, (b) intangible products in the virtual realm, and (c) the interconnections of data and information that link the virtual and physical products. The three-dimensional DT model, initially conceptualized by Grieves, is widely utilized. However, as the application demands continue to expand and evolve, the development and implementation of DT are encountering new trends and expectations. Building upon the foundational work of previous studies [3], Gupta et al. have extended the traditional four-module framework of data input, data curation, simulation, and application services into a more nuanced hierarchical structure. Their research proposes a five-layer architecture comprising physical, communication, data, intelligence, and application layers [26]. This development enriches the conventional model by integrating a dedicated intelligence layer that enhances the system's analytical capabilities and decision-making processes. Similarly, Tao et al. introduced the five-dimensional DT model to expand the use of DT in many domains, including physical entities, virtual models, services, DT data, and connections [27]. To effectively apply DT technology, Jeong et al. proposed using a DT implementation layer paradigm. They suggested that the implementation layer model pertains to the developmental stage of deploying, managing, and maintaining DT technologies. The suggested layer can implement DT technology gradually, including digital virtualization, DT synchronization, modelling and simulation, federated DT, and intelligent DT service [28]. Moreover, Soori et al. explore the integration of advanced technologies such as geospatial analytics, Building Information Modeling (BIM), machine learning (ML), knowledge graphs, and digital modelling [29]. These technologies are often incorporated into DT architectures to enhance functionalities like 3D urban modelling, sensor networks, IoT platforms, cloud databases, analytics engines, and visualization interfaces.

We will apply the concepts above to streamline the operational levels of the DT architecture that are pertinent to our review. The scope of our investigation is limited to five clearly defined functional layers: Layer one is tasked with disseminating and acquiring data. The system's second layer focuses on storing and managing data. The third layer includes data modelling and simulation. Layer four concentrates on the representation of data through visual means and ensuring its accessibility. Layer five involves ensuring data security and providing intelligent services.

2.3 Smart Cities and Related Subdomains

The concept of Smart Cities could be linked back to the emergence of wired cities during the mid-1980s [30], after digital cities by the beginning of the 1990s [31], and later the smart growth initiative [32]. The smart city notion aims to improve residents' high standards of life by establishing an advanced system that involves local governments and citizens in decision-making [33]. A city can be considered intelligent if it complies with regulations and effectively achieves planned goals. Several modelling techniques have been suggested to gain a deeper comprehension of the practical implementation of this notion, among which Giffinger's approach has attained the highest level of popularity. This approach views a smart city as a combination of six characteristics, including smart economy, smart mobility, smart environment, smart people, smart living, and smart governance. Each attribute comprises several criteria, contributing scores to evaluate cities and determine their position in the SCs index. It is crucial to balance these attributes to provide inhabitants with a good standard of living [34]. Other scholars have built on this foundational model to highlight additional components like smart infrastructure and technology [35]. Extensive studies have also investigated measurements and indices to evaluate and compare cities based on their level of "smartness" [36]. Considering the content and scope of the final remaining literature, we slightly modified the subdomains in the subsequent smart city frameworks used for this systematic review (Table 1).

3 Methods

Research on UDTs is now in its early stages, and only a limited number of research have examined their application in smart urban settings. This study highlights the prospective benefits, challenges, and objectives for forthcoming studies that combine digital technologies to augment urban efficiency, environmental responsibility, and citizen welfare. We utilized comprehensive and varied methods to examine the existing research on UDT technology and its implementation in SCs. Our process was developed based on the PRISMA-ScR recommendations [37]. Figure 2 depicts the four main phases of this systematic review: (1) identification, (2) screening, (3) eligibility, and (4) inclusion. We adhered to the PRISMA criteria since they are widely recognized as evidence-based principles that aim to ensure a transparent and full report of systematic reviews.

This review study employed the popular and renowned online libraries and electronic databases, including Web of Science (WOS) and Scopus, to search for relevant literature. These platforms were chosen for their widespread usage, reputation, and significance in information technology and urban studies. Additional resources were gathered via

Table 1. Smart city and related subdomains.

Aspects of SCs	Description of Subdomains
Smart Governance	• Political strategies and perspectives • Transparent governance and privacy protection • Public and social services • Collaborative governance and decision-making
Smart Environment	• Sustainable energy and resource management • Green infrastructure and sustainable building • Pollution monitoring and control • Environmental protection
Smart Mobility	• Local and (Inter-)national accessibility • Sustainable, innovative, and safe transport systems • Availability of ICT-infrastructure • Smart parking and traffic management
Smart Economy	• Innovation and entrepreneurship • Digital infrastructure and connectivity • Job creation and economic development • Productivity and labour market
Smart Living	• Quality of life and well-being • Health conditions and housing quality • Public safety and risk management • Touristic and recreational opportunities
Smart Society	• Digital Inclusion and ethnic plurality • Cultural creativity and social innovation • Affinity to lifelong learning • Citizen engagement and participation

Google Scholar. The query was formulated using the Boolean logic of combining the terms (“digital twin”) AND (“urban design” OR “urban plan*” OR “smart cit*”). The search terms were customized to align with the syntactic criteria of each database. The search process was conducted on titles, abstracts, and keywords spanning January 2000 to June 2023, as DTs have grown in interest since 2014 (Fig. 2). A total of 831 papers written in English were found through the searches, with 261 items identified in WOS and 570 articles in Scopus. After removing duplicate items, 606 publications were found in the original search.

Following the preliminary investigation, we established specific criteria for determining which publications would be included or excluded from the evaluation. The inclusion criteria encompassed three key aspects: (1) publication in peer-reviewed journals, (2) empirical nature of the studies, and (3) availability of the full-text version. The exclusion criteria encompassed three conditions: (1) not being pertinent to the examination relating to DT techniques within SCs, (2) being only relevant to SCs, and (3) being exclusively relevant to DT technology.

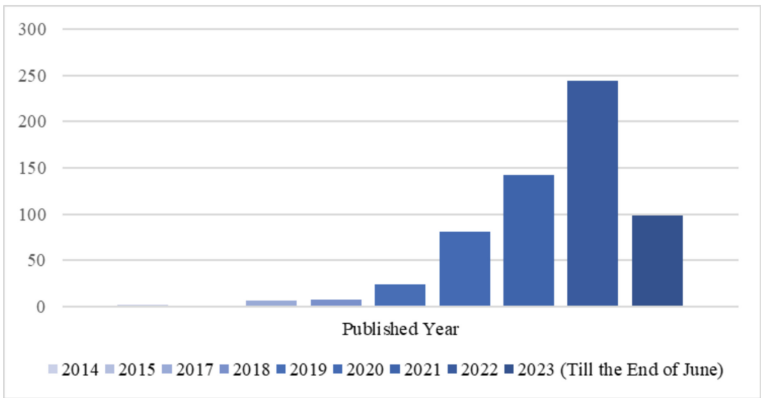


Fig. 2. Number of digital twin papers published per year.

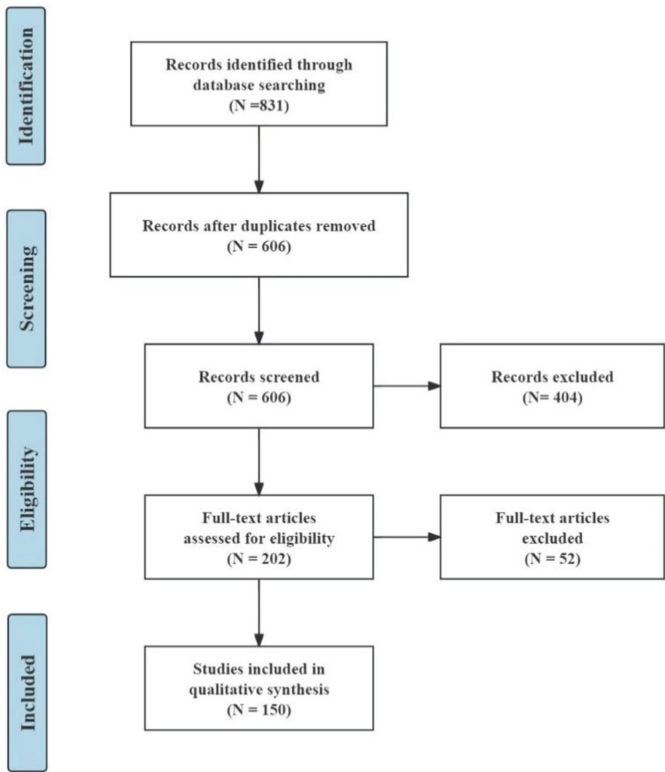


Fig. 3. PRISMA flow diagram.

The authors evaluated the papers’ titles, abstracts, and methodology sections identified throughout the initial research phase. To address prejudice, the authors actively participated in thorough conversations throughout the procedure until a consensus was

reached. The screening process involved several steps. Initially, 225 duplicate papers were eliminated, resulting in 606 remaining articles. Subsequently, 349 articles were excluded as they did not pertain to SCs, focusing instead on DT technologies or lacking detailed information on DT technology and focusing on city-level studies. Lastly, 55 articles were removed due to being review articles that did not report empirical studies or exhibited low quality based on the full text. Following a process of classification and administration, MAXQDA was utilized to import the 150 articles retained for assessment (Fig. 3).

4 Results and Discussion

4.1 Worldwide Distribution of Urban Digital Twin Applications by Countries

Urban systems exhibit deep complexity due to interdependencies and human dynamics. This requires a whole-of-city approach integrating multi-domain data at finer spatial and temporal scales [38, 39]. The implementation of DT concepts has increased significantly in recent years within the urban planning and SCs domains. Figure 4 showcases the applications of smart UDTs that are now being developed in different countries worldwide.

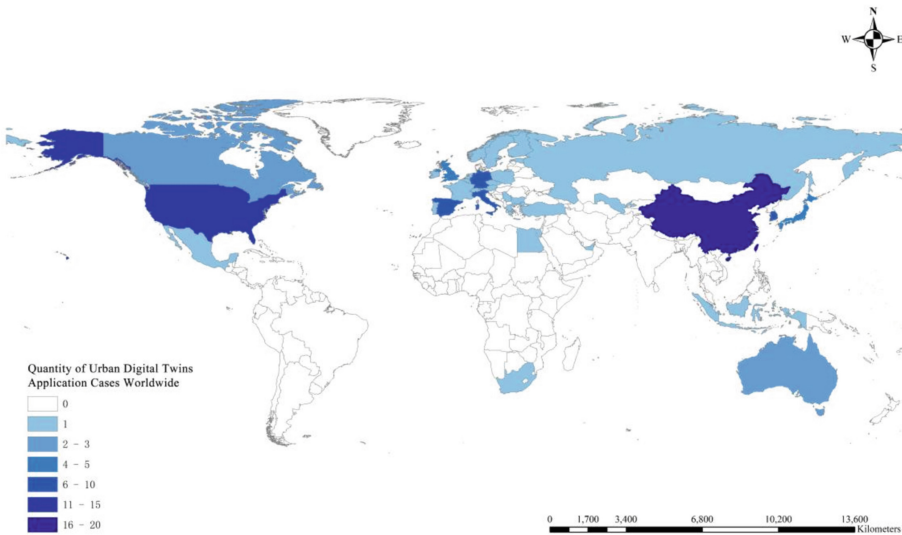


Fig. 4. Quantity of digital twin applications in different regions from the selected literature.

The adoption of DT technology is observed in 35 countries, spanning six continents. Nevertheless, the extent of implementation varies considerably. At the forefront is China, leading a notable 19 applications. This indicates a substantial investment in and prioritization of DT technologies within its industries and urban development initiatives. The United States also shows a keen adoption of DT technology with 14 applications,

highlighting its role in pioneering and integrating innovative technologies within diverse domains. Italy, Germany, Spain, and the UK, which are European nations, have adopted the technology with each country having 4–9 specific applications. Adoption in Asia is propelled by South Korea and Singapore, with 8 and 3 applications, respectively. Japan and Indonesia have spearheaded adoption in the east with 5 and 1 implementations. United Arab Emirates and Turkey represent the Middle East, while South Africa and Egypt lead in Africa. Australia, Canada, and Mexico comprise the list for the Americas list. It should be noted that 47 instances are not shown on the figure and are categorized under “N/A” due to the specific locations not being disclosed. The statistics revealed concentrated DT development in the USA, select European countries and China. Early-stage adoption is visible in Asia, the Middle East, Africa, and South America. However, many regions have yet to pilot DT projects. The data highlights the immense scope for proliferation across industrial sectors worldwide. While some countries are leading the way in DT applications, the global footprint suggests a widespread recognition of the value of this technology. On the other hand, the distribution also highlights the disparity in technological adoption, which may be influenced by factors such as economic capability, technological infrastructure, and strategic prioritization of digital transformation initiatives.

4.2 Author Keywords Co-occurrence and Thematic Analysis

VOSviewer enables the creation of a knowledge domain of DTs within the SCs architecture by analyzing the frequency of author keywords in chosen papers. The established network is structured according to distance, with the proximity of nodes reflecting the level of interconnectedness between the terms [40]. The VOSviewer clustering technique utilizes different colours to represent multiple groupings of things, while the line width reflects the magnitude of each association [41]. The four other colours depicted in Fig. 5 correspond to four distinct clusters of keywords, which are presented as follows:

Yellow Cluster: Modeling and Optimizing Urban Transportation Systems. The theme emphasizes using real-time data and simulation to develop dynamic models and extract insights from various transportation data sources. This approach aims to improve networks’ efficiency, adaptability, and ecological sustainability, facilitating a comprehensive analysis of mobility patterns, infrastructure, and planning decision-making processes. Primary uses encompass predicting traffic patterns, congestion, and travel durations by analyzing past data and real-time sensor inputs [42, 43]. Virtual modelling facilitates testing infrastructure modifications, such as introducing new transit lines, roads, crossings, or traffic management schemes. DT platforms facilitate fleet management for public transport, logistics, and shared mobility services by monitoring assets and forecasting demand.

Green Cluster: Urban Energy Management and Sustainability. The theme focuses on utilizing UDTs throughout the energy value chain, encompassing improvements in demand-side efficiency, optimization of the supply mix, and tracking sustainability. Urban innovation and energy efficiency are closely related. The predominant emphasis lies in utilizing building energy simulation and modelling to assess the effectiveness of efficiency improvements in single buildings or across districts or cities [44]. DTs

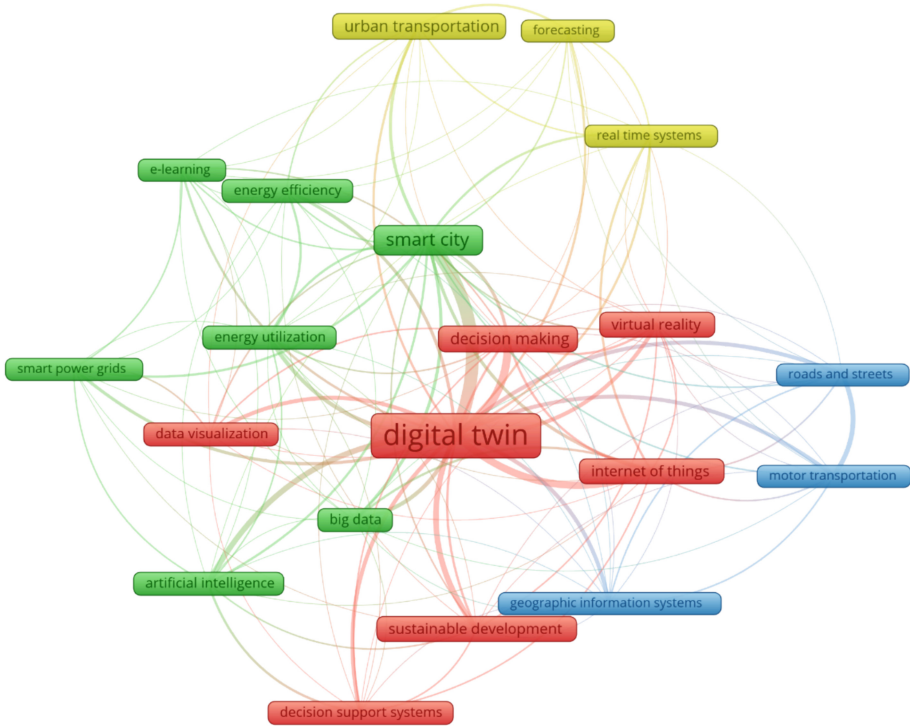


Fig. 5. Thematic clusters visualization of the keyword co-occurrence.

enable the application of predictive analytics to analyze energy consumption patterns using data on usage, occupancy characteristics, and weather conditions. This allows the detection of opportunities for optimizing heating, ventilation, air conditioning (HVAC), and lighting systems. The investigation of smart grid operations through DT analysis is extensively researched to incorporate distributed renewable production, such as rooftop solar. Additional smart city uses of DTs encompass monitoring and comparing city-wide energy consumption, greenhouse gas emissions, and sustainability indicators. The study utilizes extensive data from IoT devices and applies an ML algorithm to detect trends and generate predictions [45].

Red Cluster: Collaborative Decision-Making and Stakeholder Engagement. VR and AR technologies generate immersive settings, enabling citizens and planners to see and engage with urban digital replicas [46–48]. Visualization technologies are utilized to depict intricate urban data using three-dimensions dashboards, online maps, and performance indicators to officials and citizens. DT models offer decision support functionalities, including scenario analysis, effect assessment, and simulations of strategic initiatives. This facilitates the evaluation of planning plans, infrastructure projects, policy reforms, and other initiatives about sustainability, resilience, and equality objectives

based on empirical facts. The growing significance of DT platforms in collaborative planning and governance highlights the shift towards data-driven, transparent, and inclusive decision-making aided by advanced simulation, IoT, and visualization technologies.

Blue Cluster: Geospatial-Enhanced Transport DTs. The digital depiction of multi-modal transit networks, encompassing roadways, bike lanes, walkways, and other elements, forms the fundamental basis for city-scale simulations. Geographic Information Systems (GIS) offers specialized tools for managing, analyzing, and visualizing spatial data to meet transportation planning requirements. The primary areas of interest involve analyzing transportation demand, identifying gaps in accessibility, and pinpointing regions of congestion to guide decisions regarding infrastructure investments. GIS-based DTs facilitate the evaluation of the environmental effects of transportation, including pollution, noise, and heat island phenomena, using simulation and geospatial analysis. Geosimulation modelling combines transport and land use planning by analyzing mobility patterns and urban expansion. Utilizing real-time and past geographic data enables exploring more applications, including traffic forecasting, navigation, and logistics [49, 50].

4.3 Distribution of Publications by Application Areas of Smart Cities

From 2018 to mid-2023, the quantity of publications published in various subdomains of smart cities demonstrates a significant rise in research concerning Smart Environment and Smart Mobility, suggesting that these fields have become well-established focal points (Fig. 6). Domains such as Smart Living, Smart Society, Smart Economy, and Smart Governance have a limited but steadily growing number of publications, suggesting emerging areas that demand further research.

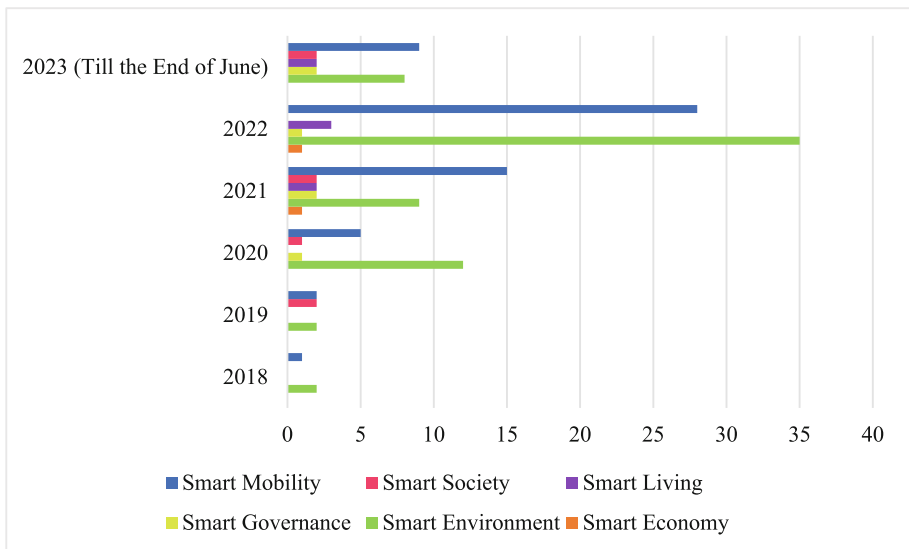


Fig. 6. Trends in SCs research by subdomain (2018 – Mid-2023).

Smart Governance. Virtual models serve as dynamic planning tools that enable city planners, administrators, and citizens to engage with a city's data more effectively and informally. By integrating spatial data, IoT, AI, and visualization technologies, DTs are redefining the urban planning and governance landscape, paving the way for more responsive, sustainable, and participatory cities. In Daejeon, South Korea, augmented reality technologies such as HoloLens are used to navigate and visualize DTs, offering stakeholders a more interactive and user-friendly experience [51]. Meanwhile, Bujari et al. proposed the IPPODAMO model in Bologna, Italy, a proof-of-concept DT incorporating a multi-layer geographical information system that integrates heterogeneous urban data to predict and quantify urban activity levels. This model supports city administrators in planning and scheduling maintenance operations by providing a consolidated view of descriptive and predictive urban data [52]. Zurich's approach extends its existing 3D city model to a more comprehensive DT, adding new data dimensions to enrich the models and enable real-time connections. This advanced model assists in urban planning, allowing simulations of scenarios such as climate change planning and high-rise building development, thereby providing a proactive tool for managing urban growth and environmental challenges [53].

Smart Environment. By enabling the co-simulation of different components and the testing of control strategies, DTs help optimize the operation of these complex networks [54]. This capability is crucial for increasing power production, reducing maintenance costs, and improving operational efficiency, as it allows for the real-time optimization of turbine operating parameters and provides insights that can inform the design and operation of future energy systems. Similarly, Huang et al. demonstrated how optimizing the configuration and operation of generators, chillers, and boilers through a DT reduced costs and energy losses [45]. For buildings, DTs integrating IoT sensor data, energy simulations, and machine learning provide capabilities for real-time energy benchmarking and identifying efficiency opportunities compared to baseline models [44]. At the city level, Gürdür Broo et al. created a 3D city model for Helsinki that serves as a visual tool, a platform for collaboration between city planners and residents, and an analytical system that combines open data from many sources. By enabling the simulation of different scenarios, stakeholders can predict the impacts on urban factors like traffic, energy use, and noise pollution before actual implementation, thus optimizing decision-making processes [55]. In water systems, "living" DTs using operational models and real-time data from sensors, meters, and SCADA systems help utilities improve water network monitoring, control, and planning. They provide decision support for daily operations and long-term supply management [56].

Smart Mobility. DTs enable advanced modelling, simulation, and optimization of transportation systems and mobility patterns in smart cities, which catalyzes traffic management, reduces congestion, and improves overall transportation efficiency [57]. For traffic management, Aguiar et al. put forward a DT of an urban road network in Alabama, which allowed testing and comparing various sustainable routing strategies through simulation before implementation. It helped planners understand trade-offs between congestion, emissions, and travel time [42]. At a regional scale, DTs combining real-time sensor feeds, traffic microsimulations and machine learning have shown promise for real-time monitoring, congestion prediction and adaptive signal control of vehicular networks

[43]. The DT framework provides a virtual proving ground, where scenarios too risky or rare to replicate on actual streets can be tested extensively [58]. DTs help optimize operations for public transit by virtually testing changes before physical deployment. The metro system DT developed by Wang et al. in Shenzhen, which incorporates IoT data, simulations, and optimization algorithms, yielded valuable insights resulting in an 11.5% reduction in energy losses [59]. DTs also enable intelligent maintenance. A DT architecture detected subway equipment faults over 10% more accurately by fusing visual, audio and sensor data than using logs alone [60]. Besides, DTs also help evaluate new transportation modes before launch. ElSayed et al. optimized solar-powered charging infrastructure for drone deliveries in Zurich by virtually modelling flights and energy dynamics [61].

Smart Economy. The potential of DTs in this domain is demonstrated by their capacity to coordinate intricate logistical processes, stimulate well-informed decision-making, and enhance the sustainability and efficiency of the construction ecosystem. DT technology is not limited to robots and is also being extensively used in the construction sector within the on-site assembly of precast materials. Jiang et al. delineate using DTs for real-time synchronization of planning, scheduling, and execution (PSE), wherein smart construction objects equipped with IoT sensors feed continuous data into the cloud-hosted DT. This real-time data flow engenders unprecedented visibility and traceability, empowering stakeholders to precisely monitor construction progress and resource status [62]. The DT becomes an intelligent nexus for tracking and dynamically adjusting schedules and task allocations, optimizing construction workflows and bolstering project resilience. Despite the transformative potential of DTs in the Smart Economy, their deployment is challenging. Reliability of data, as highlighted by Tennakoon [63], remains a pivotal concern, as the effectiveness of a DT is inherently tied to the accuracy of the sensor data and the robustness of the communication infrastructure. The simulation fidelity and the capacity to generalize models across varied operational conditions also present significant hurdles that necessitate further research and innovation. Future directions for DT technology in the Smart Economy include enhancing model reliability through sensor fusion techniques, broadening the application scope to incorporate forecasting and optimization, and integrating insights from DTs directly into physical system controls. The real-time data synchronization between the digital and physical domains fosters enhanced planning and execution of construction tasks, leading to a more resilient and adaptive construction process.

Smart Living. DT technology, a staple of the Industry 4.0 revolution, has permeated smart living by introducing innovative ways to simulate, monitor, and manage the complexity of urban and living environments. In smart living, a DT serves as a dynamic, digital counterpart to physical spaces and entities, integrating real-time data, predictive analytics, and intelligent decision-making to enhance the quality of life, streamline city services, and foster sustainable urban development [64]. A DT framework integrates BIM with real-time sensor data from equipment and components on a construction site in the building and construction industry. The virtual representation of the site enables remote visibility, coordination, safety evaluations and quality control during construction [65]. Agostinelli also suggested introducing DTs for entire city information systems, combining BIM, GIS, IoT data, computer vision and AI on digital platforms [66]. This

allows automated security surveillance, predictive maintenance, and resilience through city-scale DTs. Chio et al. developed the SmartSPEC DT platform for smart spaces, which involves creating a comprehensive virtual model that encapsulates every aspect of smart space, such as spatial configurations, resident behaviours, and event occurrences [67]. This model is continuously fed with data from various sources, including sensors that track environmental conditions and human interactions, to simulate the living environment robustly. This facilitates testing IoT applications without needing real user data, overcoming privacy concerns. DTs also enable evaluating hypothetical emergency scenarios and visualization for space planning. In real estate, DTs of apartment projects allow remote walkthroughs, customizations, and transactions for potential buyers. Real-time integration of unit availability, weather and other data sources enhances accuracy – game engines and VR aid virtual experiences [64]. At a device level, network DTs help improve the security and trustworthiness of IoT systems. A smart building example simulated IoT devices and network traffic to test security frameworks and behavioural anomaly detection before deployment [68]. By replicating devices virtually, DTs enable testing resilience against cyber-attacks and conflicts between device behaviours and security policies.

Smart Society. By creating virtual replications of cities and public spaces, DTs allow collaborative evaluation of urban planning scenarios. This signifies a significant change in how urban environments are created, controlled, and encountered. For instance, Wang and Vu highlight the integration of DTs with serious gaming frameworks, using Minecraft to create a virtual prototype of the Dadaocheng area in Taipei [47]. This prototype is a tool for urban exploration and social interaction, enabling users to engage in collaborative urban planning through an interactive 3D virtual environment. Through immersive VR environments and interactive visualizations, citizens and experts alike can engage with and contribute to the intricate processes of urban planning, exploring various development scenarios and evaluating their potential impacts on the urban fabric. Furthermore, the literature emphasizes the multifaceted technology stack that underpins the development of DTs. This includes 3D modelling and laser scanning to recreate urban architecture and infrastructure accurately, applying space syntax theories for analyzing movement patterns within urban street networks and deploying urban mobility simulations to visualize and predict traffic dynamics. At a neighbourhood scale, a DT in the Netherlands allowed inhabitants to evaluate and customize proposed housing block designs early – this enhanced transparency and citizen inclusion in municipal decision-making. By bridging the gap between physical and virtual worlds, DTs serve as an integrative approach that not only improves the visual and interactive qualities of the DT but also ensures that the model remains an accurate and up-to-date reflection of the physical counterpart, allowing stakeholders to test and analyze the implications of various urban planning decisions before they are implemented.

4.4 Enabling Technologies in Functional Layers of UDTs Architecture

This study classifies technologies supporting DTs in SCs based on the five layers of a general UDT architecture discussed in Sect. 2.2. During the data management procedure, DT collaborated to develop a sophisticated system with multiple layers by incorporating

various technologies into each layer. The DT platform not only replicates and mirrors the present condition of a tangible entity but also anticipates potential futures and aids in the decision-making process. The complete process, starting from the gathering of data to the sophisticated analysis, showcases the capabilities of DT technology. By coding 150 papers, the individual technologies mentioned in the text were categorized, summarized, and refined. Table 2 provides a comprehensive overview of the technologies used in each subdomain of the smart city, as well as the five layers inside the UDT design, all of which are supported by DT technology.

Table 2. Enabling technology application in UDT-supported SCs.

DT Layer	Enabling Technology	Smart Mobility	Smart Environment	Smart Economy	Smart Society	Smart Governance	Smart Living
Data Dissemination and Acquisition	Remote Sensing	X	X				
	UAV/Drones (Unmanned Aerial Vehicles)	X	X				
	Cameras and Laser Scans	X	X		X		
	Sensors (Including Physical Sensors and IoT Sensors, AoT Sensor Nodes and Wireless Sensor Networks)	X	X	X	X		X
	Volunteered Geographic Information (VGI) / Crowdsourced Data	X	X				
	Mobile Apps and Application Programming Interfaces (APIs)	X	X		X		X
	SCADA (Supervisory Control and Data Acquisition) Systems		X				
	IoT Devices and Internet of Vehicles (IoV)	X	X	X		X	X
	Wearable Biosensors and GPS	X					
	Photogrammetry and AR/Webcam Scanning		X		X		
	Cellular Networks, 5G, Broadband Fiber, WiFi and Bluetooth		X				X
	Public Geospatial Data (GIS / BIM / CAD)	X	X		X	X	X
Data Storage and Management	OpenStreetMap (OSM) / Google Street View / Satellite Imagery		X		X	X	
	IoT Devices Integration	X	X				
	Recording Systems			X			
	Cloud Platforms		X				X
	Data streaming	X					X
	Relational Databases (PostgreSQL, Cassandra)	X	X				
	Data Fusion Algorithms		X		X	X	X
	Data Visualization and Analysis	X					
	Middleware / Digital Threads	X	X				
	Standards and Protocols (FBX, OPC-UA Protocols, ISO 37120)				X		X
	GPS and Sensor Technology			X			X
	Syntactic and Semantic Transformations					X	
Data Modeling and Simulation	Computing Infrastructure	X			X		
	Bidirectional Data Flows			X			
	Blockchain and Security	X				X	
	Traffic and Environmental Simulation (Gazebo / ANSYS/ SUMO)	X		X	X		X
	Physics-Based Models		X				
	Energy Simulation (EnergyPlus, SmartSPEC)		X			X	X
	Agent-Based Modeling / Cellular Automata (CA) Model	X					
	Semantic and Virtual Models		X	X	X		X
	Unreal Engine / Interactive Tools				X		X
	3D BIM and GIS Integration		X		X		X
	Machine Learning, Reinforcement Learning and Analysis Software	X	X			X	X
	3D Modeling	X	X	X	X	X	X
Data Visualization and Accessibility	Geographic Information Systems (GIS)	X	X			X	
	Three.js Library / 360-Degree Views	X				X	
	VR/AR Technologies (Including Headsets and Motion Controllers)	X	X	X	X	X	X
	Web Interfaces	X				X	X
	Dashboards	X	X		X	X	
	Time-Series Database			X	X		X
	Tabletop interaction and Apps	X		X		X	
Data Security and Intelligent Services	Access and Permissions Management	X	X			X	X
	Data Anonymization	X					
	Monitoring and Control			X			X
	AI/ML Integration	X	X			X	X
	Data Management and Analysis	X	X		X	X	
	Intelligent Routing and Mobile Apps	X					

5 Benefits, Open Issues and Key Challenges

Deploying DTs in urban settings provides a diverse range of advantages. These elements include enhancing operational efficiency and innovation, promoting public engagement, fostering sustainability, and implementing the DT system, which integrates accurate and dynamic models, resulting in a substantial revolution in city governance, infrastructure advancement, and public engagement. This is achieved by facilitating real-time monitoring and predictive modelling with immersive visualization in urban ecosystems. One of DTs' biggest advantages is that they improve the decision-making process. These complex urban systems are studied based on multiple data sources and act as cooperative platforms, engaging stakeholders in the development process by enabling them to access data universally [46, 48, 62, 66]. By doing so, every stakeholder can learn how their decision impacts an urban area's environmental sustainability and social development. DTs display the results of urban planning in dynamic and interactive visualizations [69, 70], showing real-time modifications to help the adaptability of urban systems to evolving needs [42, 62, 71]. For cities that want to reduce carbon emissions and achieve net-zero goals, the DTs are essential for the sustainability of cities by utilizing the platform that examines the potential effects of energy programs and environmental regulations to verify them before wide implementation.

However, the hybrid opportunity of considering DTs in the urban context is certainly diverse – but also complex in the various technical, organizational, and sociological challenges it implies. The high accuracy of measurement requirements constrains the potential of DTs. The design of high-fidelity virtual models requires the presence and combination of plenty of live data. This task is often hampered by data interoperability concerns, where different types of sensors, precision of information, and data coming from multiple sources must be investigated [44, 57, 65, 72]. Similarly, scaling up these digital replicas from small-pilot trials to wider urban, heterogenous systems heightens the risk of such challenges. Interoperability issues must be addressed with common data exchange and system integration protocols [63]. The processing capabilities increase accordingly, calling for a resilient infrastructure to crunch and render huge amounts of data in real-time.

When existing operating procedures are followed, organizations usually face opposition. Stakeholders must foster successful change management during the shift from traditional to digital and data-driven approaches [53, 70]. One such complex duty is coordinating different sources of information, such as IoT devices and feeds from satellite imagery. However, other human factors affect the implementation of DT technologies, like challenges in engaging users, different technical maturities across industries, communication gaps between cross-disciplinary teams and the need for extensive training in DT technologies [55, 73]. Additionally, it should be noted that cyber security has become a concern due to the interconnectivity of DTs and reliance on Information Technology and Operational Technology (IT/OT) convergence [57, 68]. For instance, urban systems that deliver public services and ensure safety are critical locations where sensitive data must be shielded from cyber risks. However, integrating DT technology into smart cities has several challenges and obstacles. These include issues relating to technicalities, data management, and those pertinent to humans and organizations. Thus, strategic solutions must be employed to actualize this potentially promising technology's possibilities.

6 Conclusion and Future Directions

The DT is undergoing rapid development, giving rise to numerous study forms. Simultaneously, the study of DTs is becoming increasingly in-depth and specialized. This article critically examines 150 peer-reviewed articles exploring DT technologies' fundamentals, research methodologies, and application domains. The papers were published between 2018 and mid-2023. An analysis and debate are conducted. The main contributions to this work are as follows:

- 1) This analysis examines the entire adoption of DTs in SCs to provide a clear understanding of their definitions, features, and application situations.
- 2) The analysis systematically examines the features and processes of DTs, classifying various technologies according to their implementation phases.
- 3) This study summarizes the prevalent application domains of DTs and highlights the main issues, benefits, and unresolved matters.

Despite the growing prevalence of DTs, research in this area is still in its early stages of rapid progress. Experts have suggested several DT reference architectures and created diverse DT components utilizing various implementation methods, demonstrating the extensive scope of application sectors. This highlights the necessity for more agreement throughout the corporate sector regarding the current research on DTs, which poses difficulties in carrying out systematic investigations. Hence, designing a cohesive platform for multi-domain DTs that effortlessly incorporates model development, testing, and implementation is imperative.

As we venture into the future, our vision is to extend the scope and scale of DTs, advancing from individual asset models to encompass entire urban infrastructures [44, 71, 74, 75]. The primary objective is to integrate real-time data from multiple sources [42, 73, 76], allowing for a dynamic and detailed representation of the city's pulse. To do this, one must utilize sophisticated simulation methods that use artificial intelligence technology to improve prediction skills [43, 50, 77]. Consequently, this allows for astute decision-making. Standardized platforms and architectures are essential for the future progress of DTs. They provide consistent and efficient interoperability across many domains within the SC mega system [45, 53, 63, 67]. Furthermore, the availability of immersive technologies such as AR and VR would promote transparency and engagement in urban management and planning. This will make these interfaces accessible to stakeholders and residents democratically. By effectively combining operation and technology, it transforms from a passive assessment tool to an active participant in urban management [47, 48, 51, 78]. In the future, the combination of digital technological breakthroughs, enhanced computing efficiency, diverse interaction techniques, and widespread adoption of UDT will result in the smooth integration of the digital and physical parts of urban settings.

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